Module 1

Explaining Network Topologies

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Learning Objectives

- Explain network types and characteristics.
- Compare and contrast OSI model layers.
- Configure SOHO networks.
- Explain CompTIA’s troubleshooting methodology.

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Lesson 1.1

Networking Overview

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Think About It

How would you define your social network?

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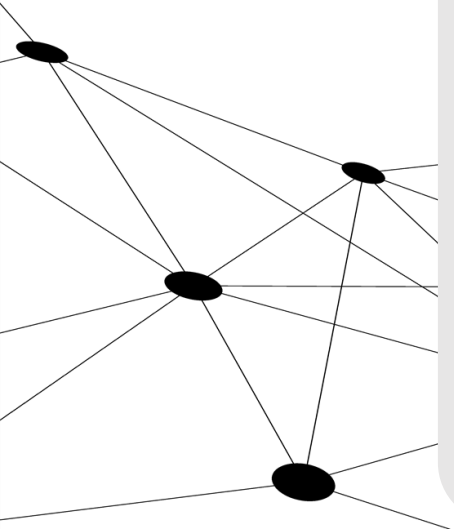
How are Personal and IT Networks Similar?



Social Network

- A collection of friends, family, coworkers, and acquaintances
- Provides emotional support, advice, and opportunities in personal matters
- Helps in communication, information sharing, and social connections

IT Network

- A collection of computers and other devices connected by transmission medium
- Provides an opportunity for shared resources
- Helps with communication and information sharing in a digital environment

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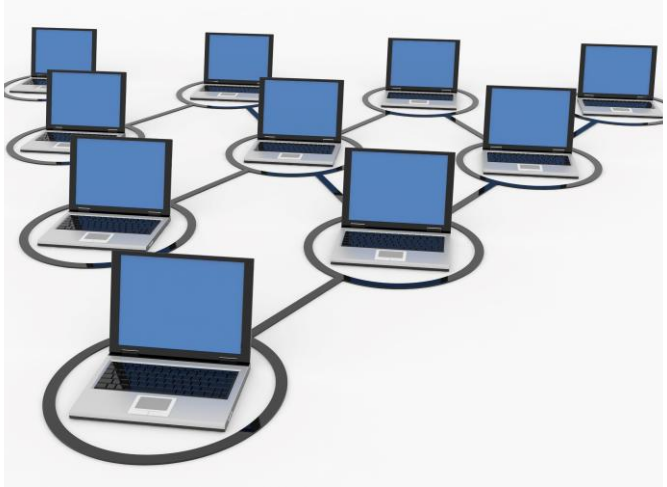
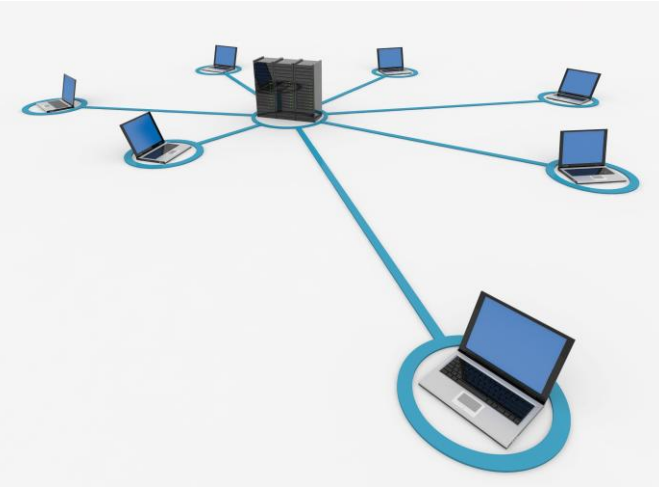
Networks



Peer-to-Peer



Client-Server

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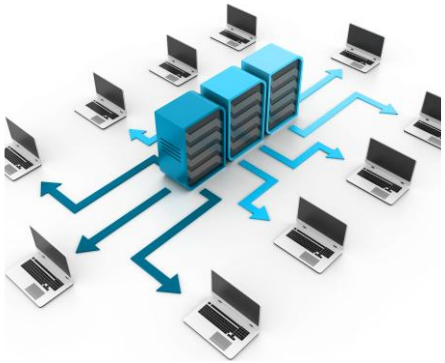
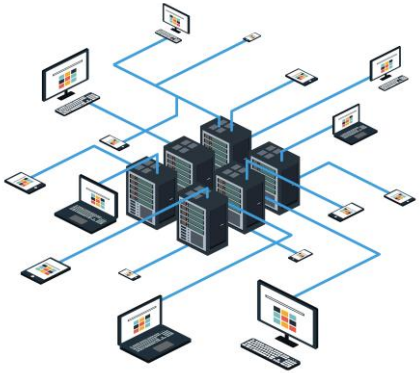
Local Area Networks

A local area network (LAN) is confined to a single geographical location.

SOHO



Enterprise LAN



Datacenter

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Activity: Quick Search

Point-to-Point Topology

Star Topology

Mesh Topology

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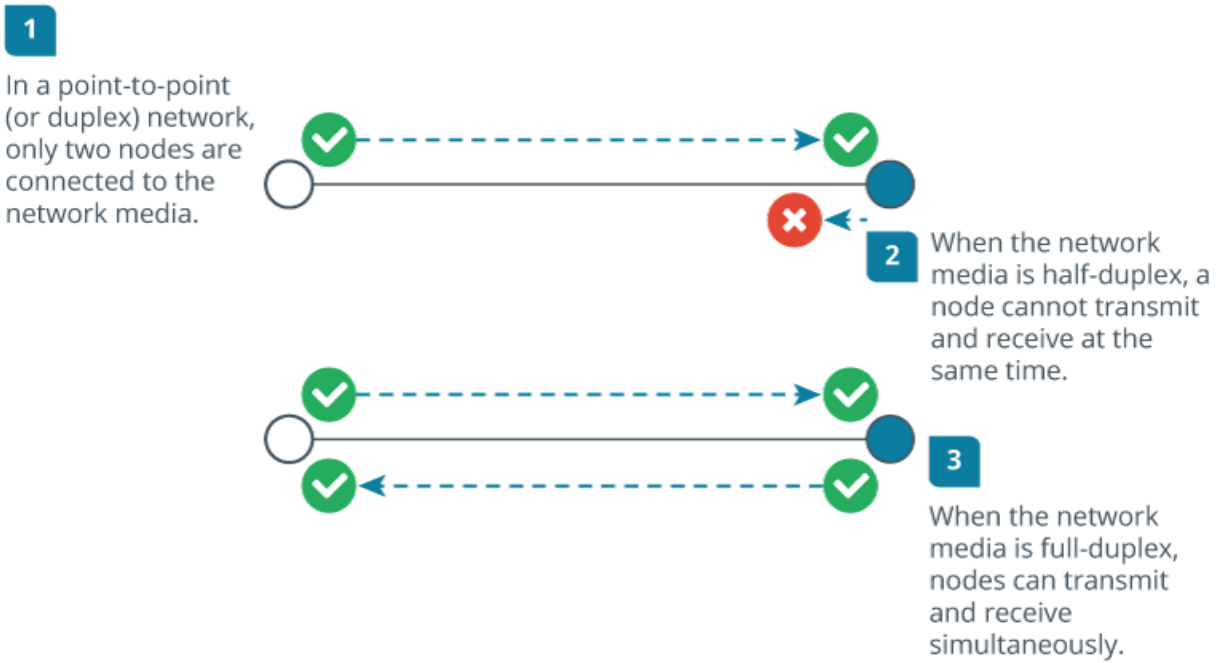
Network Topologies: Point-to-Point



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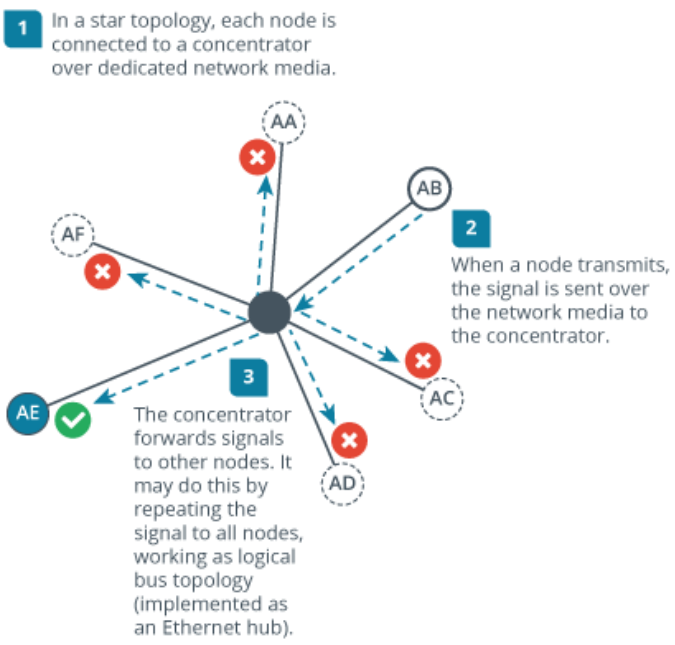
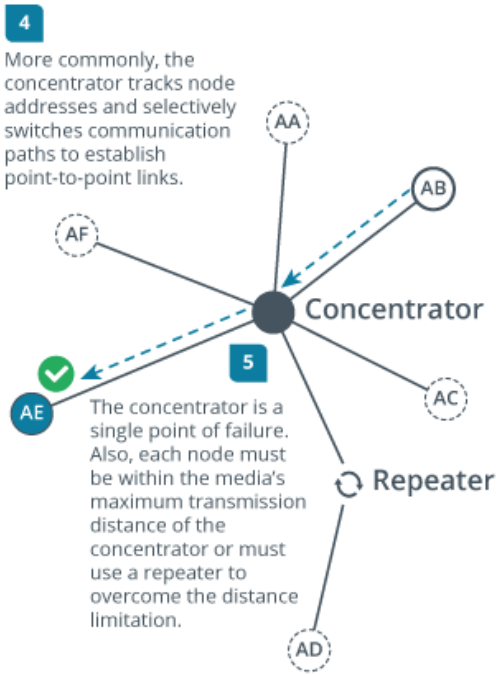
Network Topologies: Star





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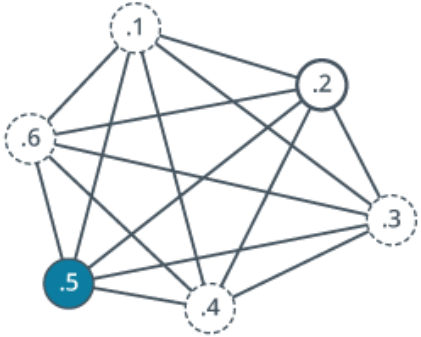
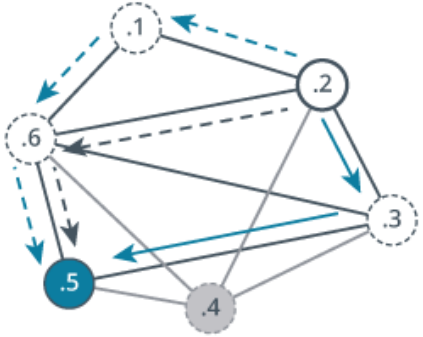
Network Topologies: Mesh

1 In a fully connected mesh network, each node has a point-to-point link with every other node. This requires exponentially more links as nodes are added: n*(n−1)/2



2 Provisioning so many interfaces and links is difficult, so partial mesh networks are often preferred, In a partial mesh, nodes can forward packets to a destination by learning the network topology.



3 Packets can take multiple routes through the network, providing resilience if some nodes or links fail.

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Lesson Summary

- **Network**: Connected devices sharing data (computers, protocols, transmission medium)
- **Client-Server Network**: Central server provides resources and services to client devices
- **Local Area Network (LAN)**: Connects devices within a limited area (home, office)
- **Network Topology**: Layout of connections between network devices (star, mesh, point-to-point)

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Lesson 1.2

OSI Model Concepts

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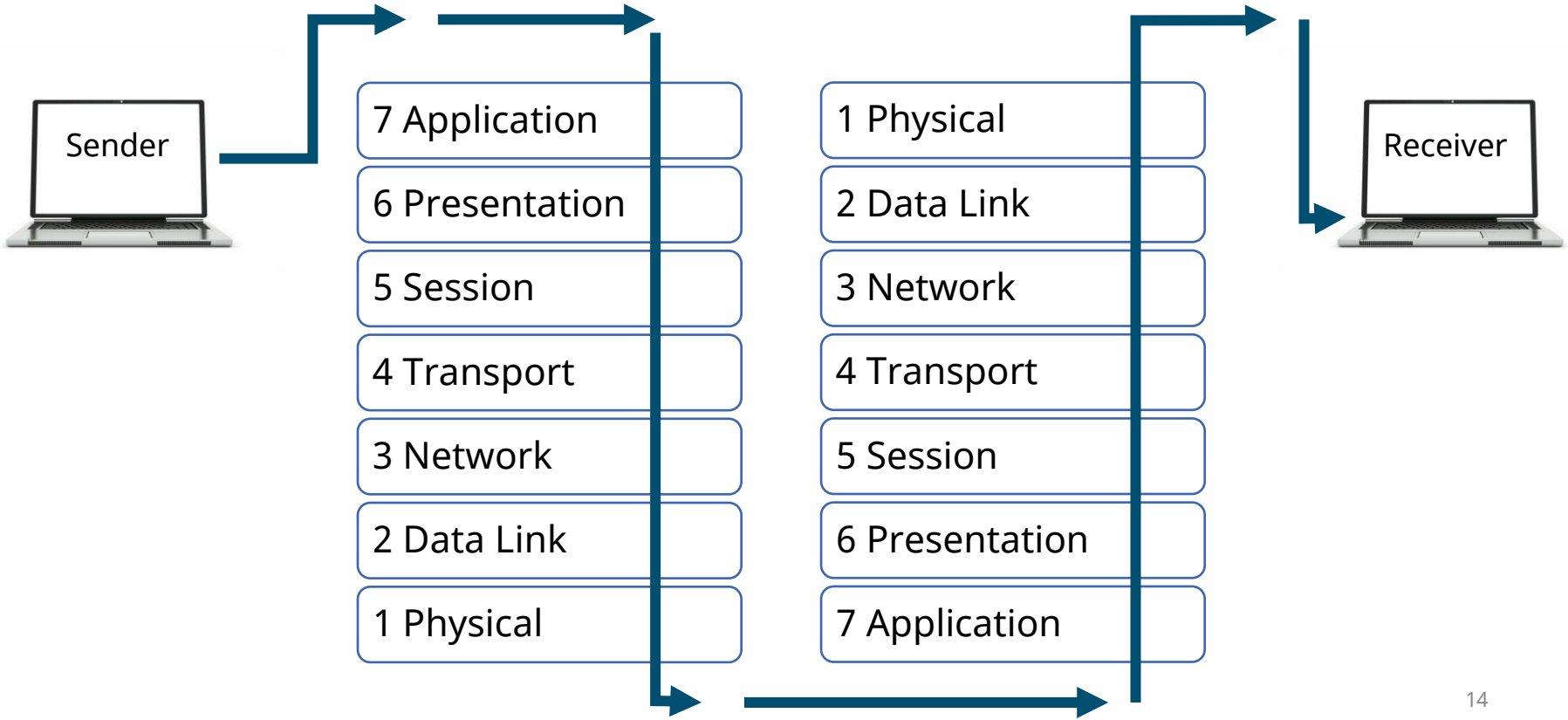
The OSI Model Packet Flow



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The OSI Model Mnemonic

Please	Do	Not	Throw	Sausage	Pizza	Away
P	D	N	T	S	P	A
Physical	Data Link	Network	Transport	Session	Presentation	Application

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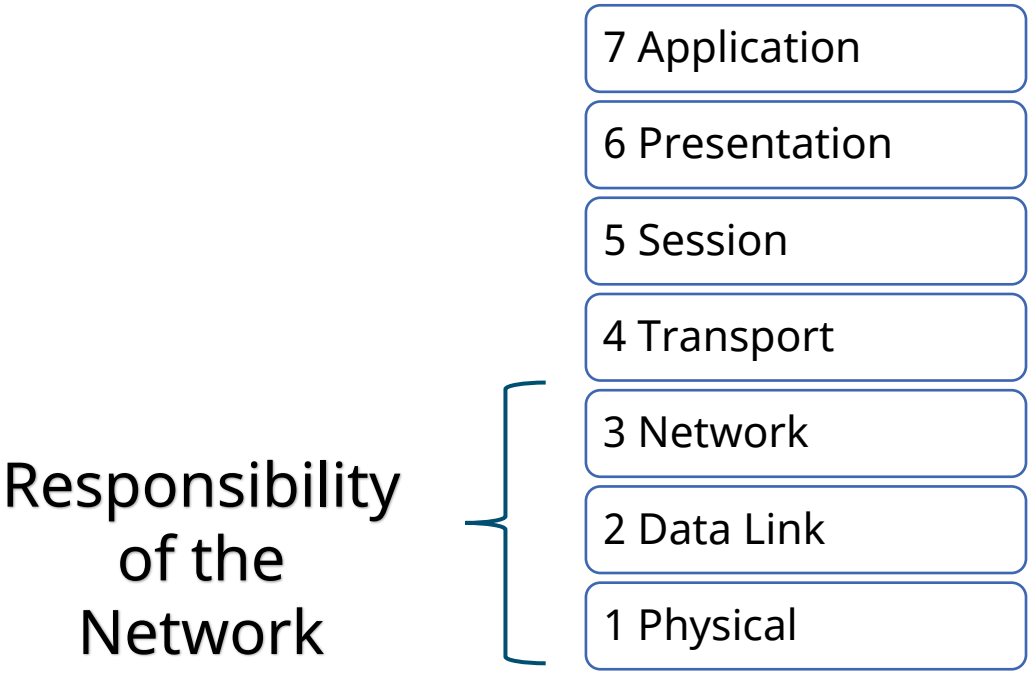
Responsibility of the Network



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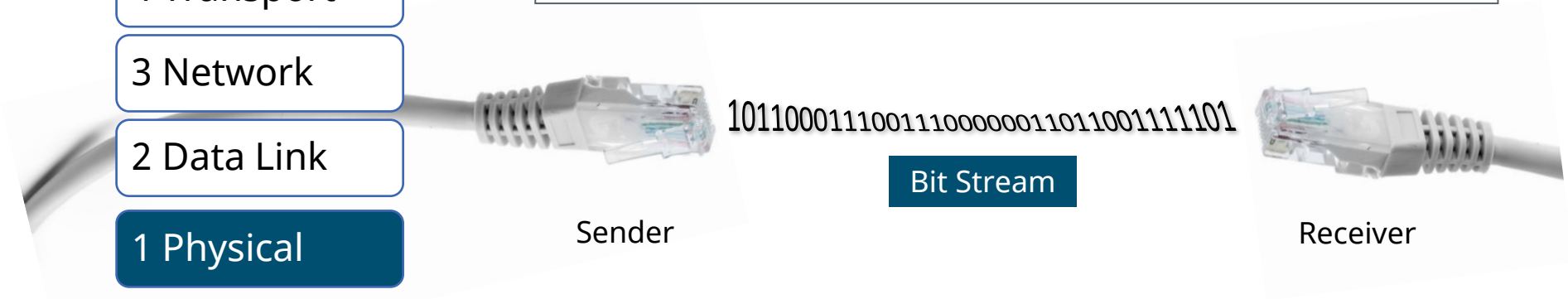
Layer 1 – Physical

CompTIA

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- **1 Physical**

Functions and Examples

- Transmit bits from sending source over network communications to physical layer of receiving device
- Data transmitted using cables or wireless media
- Examples: Coax and fiber cables, hubs and repeaters, modem, transceiver



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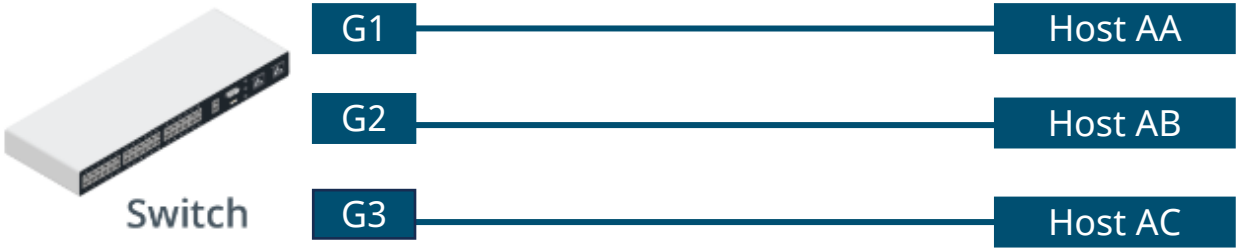
Layer 2 – Data Link

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- **2 Data Link**
- 1 Physical

Functions and Examples

- Transfers data between nodes on the same logical segment using hardware addresses
- Encapsulation: organizes stream of bits arriving from the Physical layer into frames
- Examples: NIC, bridge, switch, wireless access point



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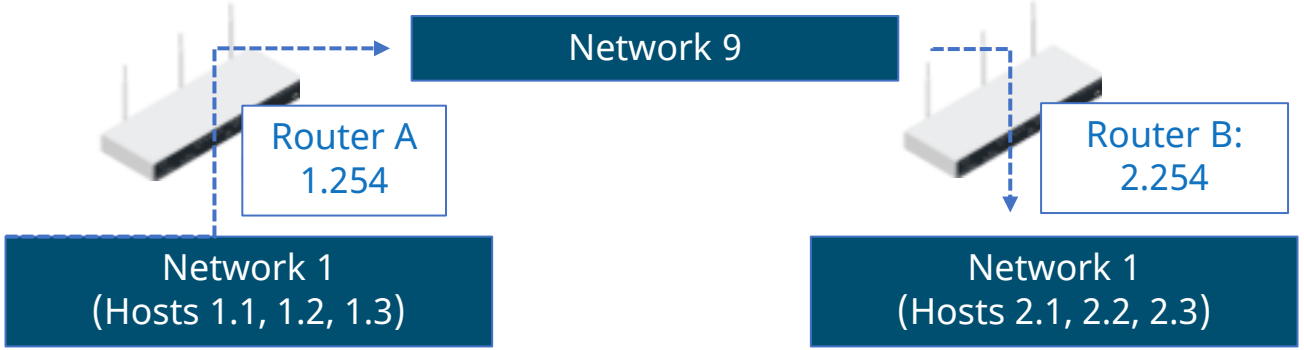
Layer 3 – Network

CompTIA

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- **3 Network**
- 2 Data Link
- 1 Physical

Functions and Examples

- Moving data around a network of networks using network and host IDs
- Packets given a destination IP address and forwarded to the destination network
- Examples: Router, IP, ACL, Basic Firewall



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Layer 4 - Transport

CompTIA

- 7 Application
- 6 Presentation
- 5 Session
- **4 Transport**
- 3 Network
- 2 Data Link
- 1 Physical

Functions and Examples

- Tracks communication between applications on source and destination hosts
- Segments data and manages each data piece
- Reassembles segments into application data
- Examples: Multilayer switches, advanced firewalls, intrusion detection systems (IDSs)

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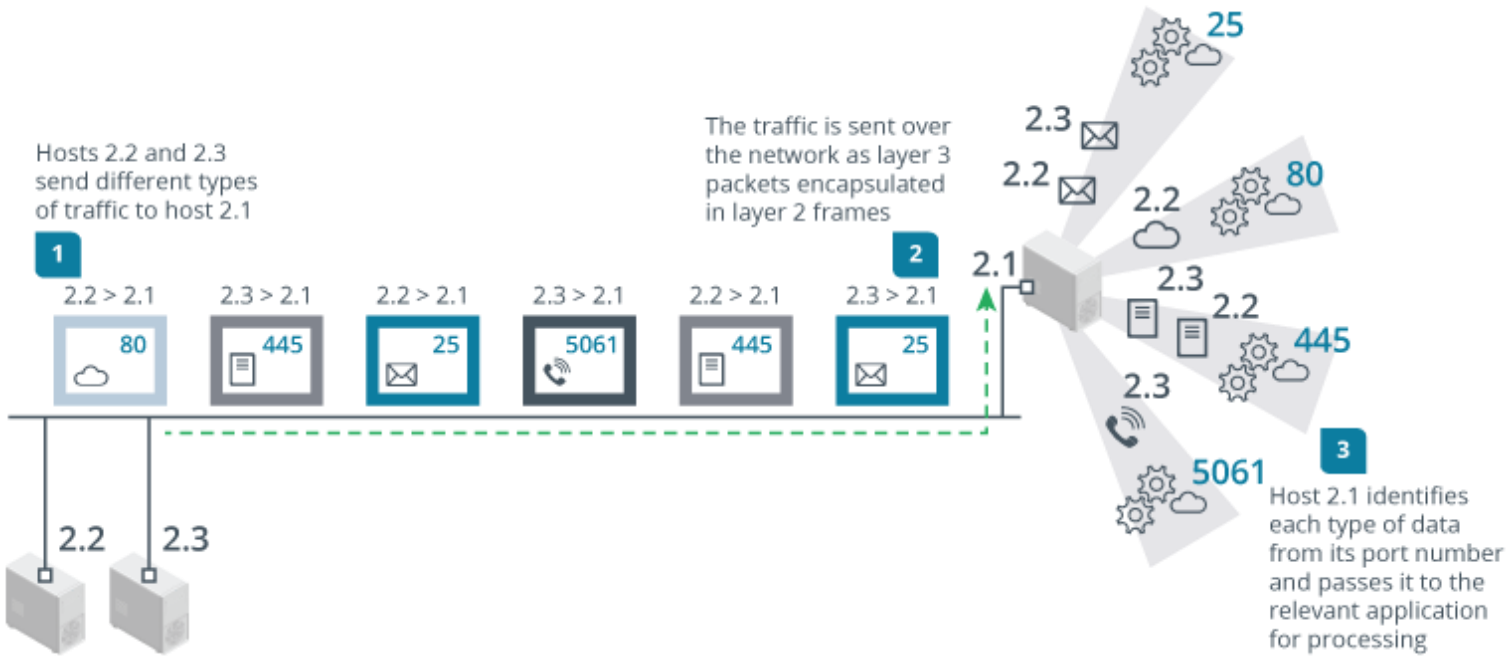
Layer 4 – Transport



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Responsibility of Host

Responsibility of the Host

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

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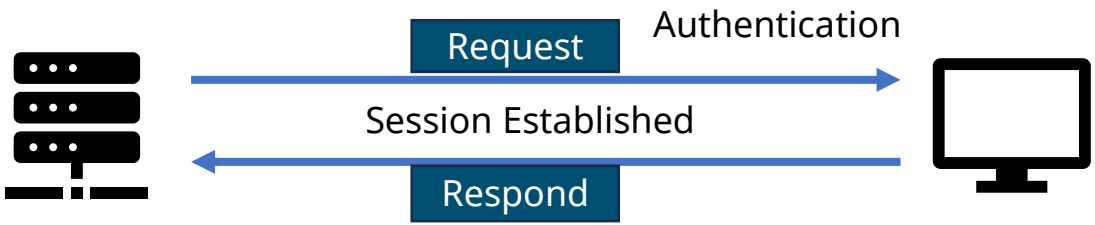
Layer 5 - Session

CompTIA

- 7 Application
- 6 Presentation
- **5 Session**
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

Functions and Examples

- Establishes and manages the connections between applications
- Examples: ASP, ADSP, NetBIOS, PAP



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Layer 6 - Presentation

CompTIA

- 7 Application
- **6 Presentation**
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

Functions and Examples

- Delivers data across network connections
- Examples: MIME, Encryption/Decryption Services

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Layer 7 - Application

CompTIA

- 7 Application
- 6 Presentation
- 5 Session
- 4 Transport
- 3 Network
- 2 Data Link
- 1 Physical

Functions and Examples

- Interacts directly with end-user's software applications for network service
- Examples: SMTP, DNS, FTP

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Activity: Unscramble the OSI Model

Network

Application

Physical

Transport

Session

Data Link

Presentation

Physical	Data Link	Network	Transport	Session	Presentation	Application

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Lesson 1.3

SOHO Networks

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Think About It

What is a SOHO router?

And what is it used for?

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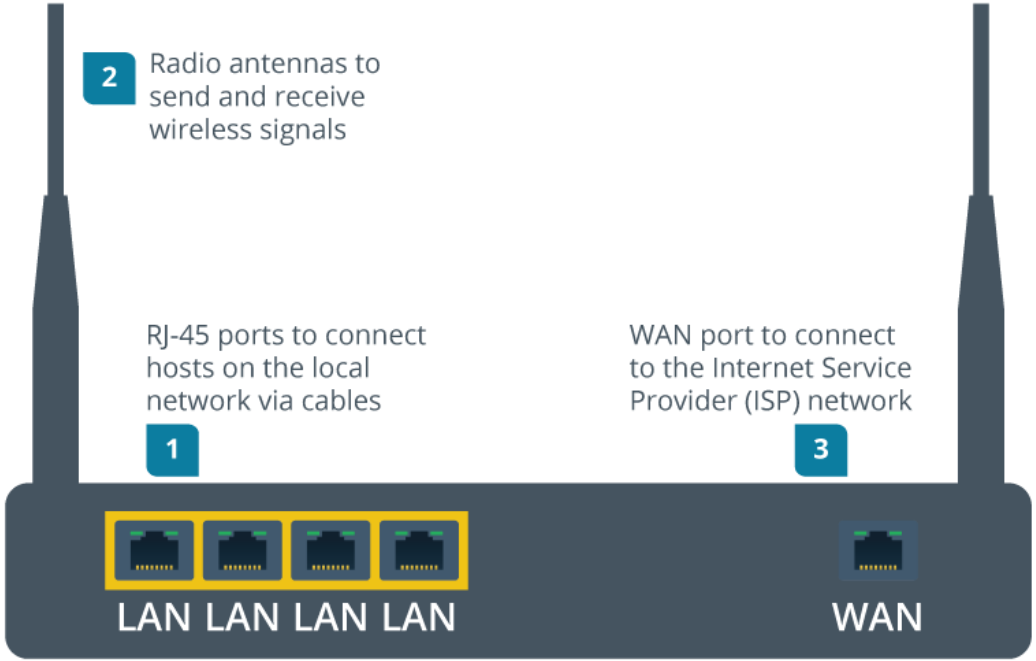
SOHO Router – Physical Layer



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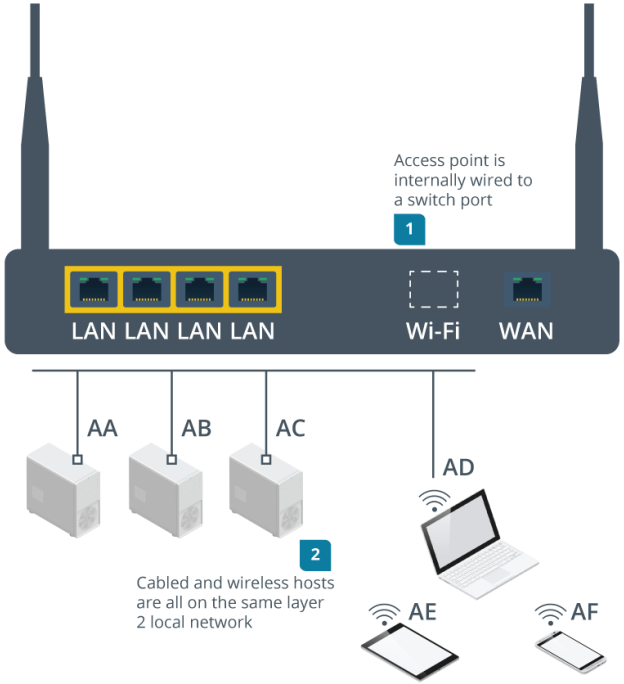
SOHO Router – Data Link Layer

- Ethernet switch
- Wireless access point
- Each host interface identified by a MAC address



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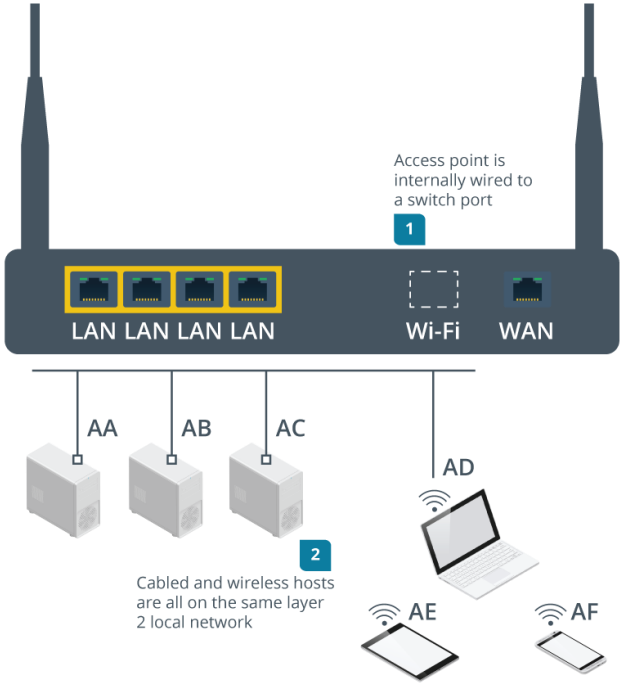
SOHO Router – Network Layer

- Forwards between local private network and public internet
- Runs a DHCP server to provide each host with an IP address
- Router's WAN interface given a public IP address



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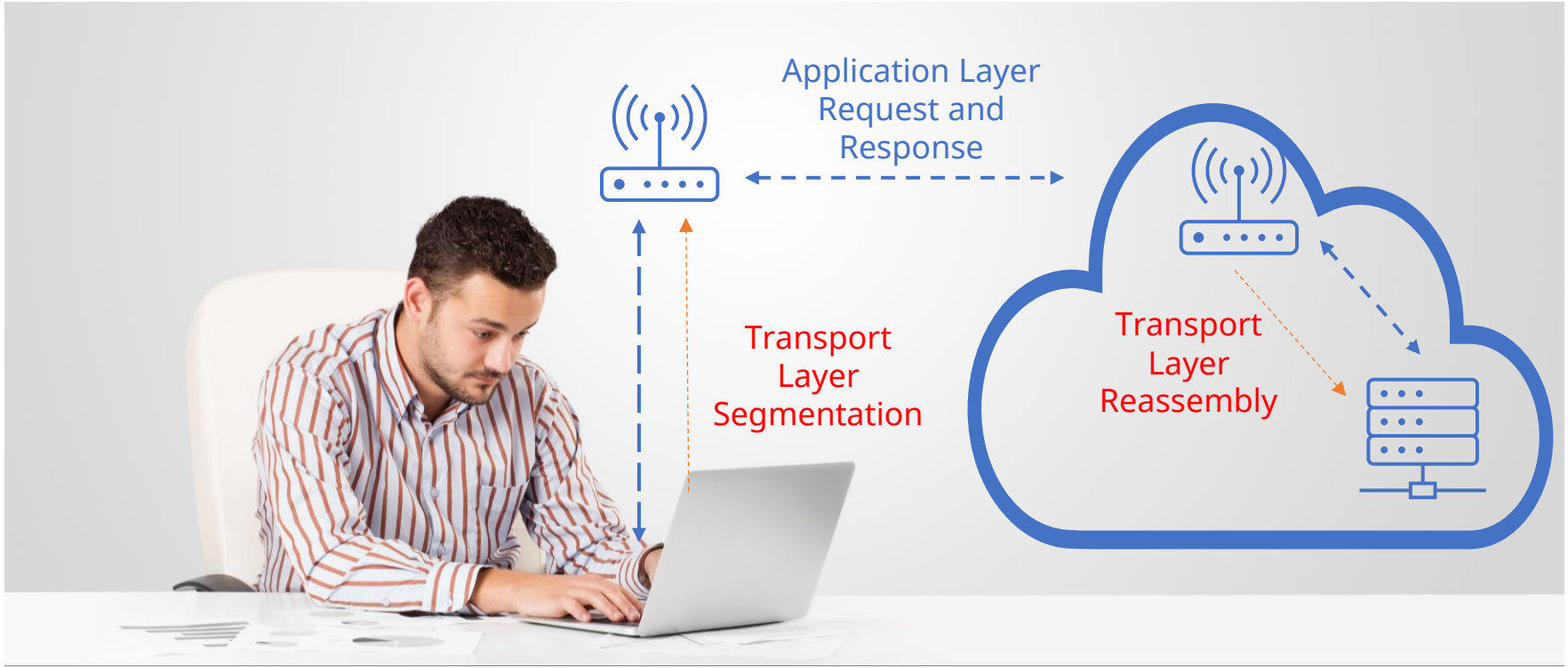
SOHO Router – Transport and Application



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SOHO Router – The Internet

- WAN: SOHO connection to Internet
- Public switched telephone network (PSTN)
- High-bandwidth trunks connecting IXPs
- ISP links between networks in IXP datacenter

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Binary Numbers – Base 10

Ten-millions	One-millions	Hundred-thousands	Ten-thousands	One-thousands	Hundreds	Tens	Ones
Digit x 10,000,000	Digit x 1,000,000	Digit x 100,000	Digit x 10,000	Digit x 1,000	Digit x 100	Digit x 10	Digit x 1

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Binary Numbers – Base 10

128	64	32	16	8	4	2	1
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1

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Binary Conversion

Let's look at how to convert 205 into binary.

128	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1
1	1	0	0	1	1	0	1

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Activity: Binary Conversion

Convert 132 into binary.

128	**64**	**32**	**16**	**8**	**4**	**2**	**1**
Digit x 128	Digit x 64	Digit x 32	Digit x 16	Digit x 8	Digit x 4	Digit x 2	Digit x 1
1	0	0	0	0	1	0	0

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Lesson 1.4

Troubleshooting Methodology

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Discussion: Think About It

What is troubleshooting?

What steps should be involved in the troubleshooting process?

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Troubleshooting Methodology

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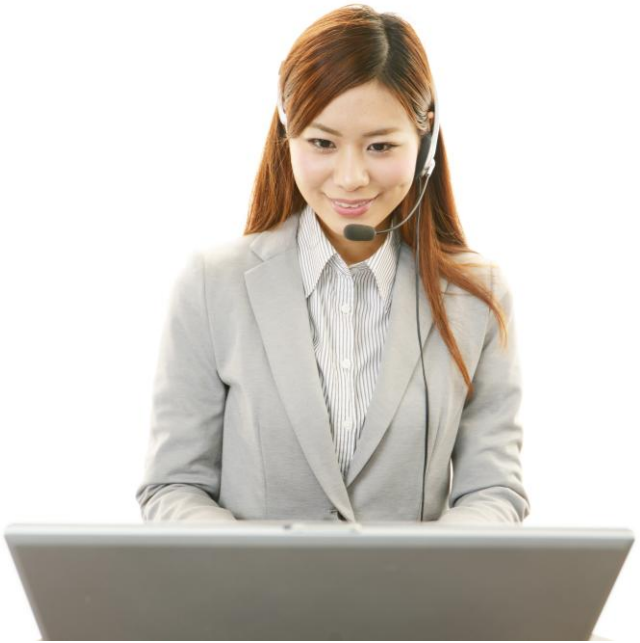
Identify the Problem



Gather information

- System documentation
- Installation and maintenance logs
- Vendor support sites

Question users

- Open questions to uncover information
- Closed question to invite a yes or no or fixed response

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Identify Problem Symptoms

CompTIA

Identify symptoms

- Physical inspection
- Logs and diagnostic software
- Try to duplicate the problem

Determine if anything has changed

- Did it ever work?
- What has changed since it was last working?

Approach problems individually

- Verify symptoms are related before treating them as one single issue

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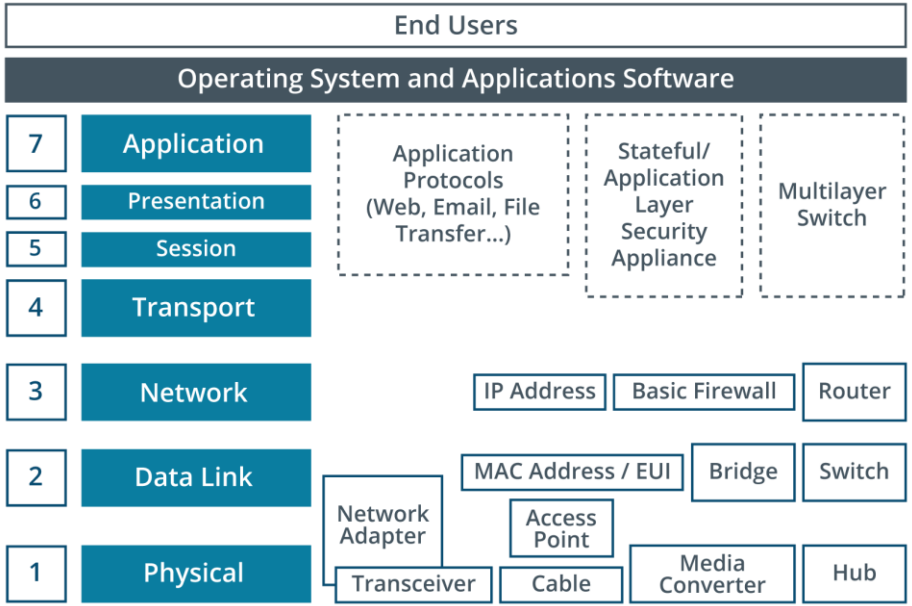
Establish a Theory of Probable Cause

Troubleshooting styles

- Question the obvious
- Methodically prove the functionality of each component

OSI Model Approach

- Top-down approach
- Bottom-up approach



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Test the Theory to Determine the Cause

Question the obvious

- Use one or more method-based approaches
- Don’t jump to conclusions

Escalation

- Problem is beyond your knowledge or ability
- Problem falls under a system warranty
- Scope of the problem is very large
- Customer is difficult or abusive

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Implement, Verify, and Document

Implement changes

- Implement fixes
- Escalate if higher authorization is needed

Verify the solution

- Make sure the system seems to work for you
- Make sure the system seems to work for customer

Document

- Document findings, actions, and outcomes
- Logs are useful for future troubleshooting

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Activity: Troubleshooting

Several office employees report their computers are unable to connect to the Internet, affecting workflow and productivity.

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Summary

- Remember OSI model’s 7 layers (physical, data link, network, transport, session, presentation, and application)
- Plan network structure and troubleshoot issues using OSI model layers
- Implement methodical approach to troubleshooting (identification, planning, execution, verification, and documentation)
- Use strategies like top-to-bottom, bottom-to-top, and divide-and-conquer

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